

Effects of salinity, irradiance and photoperiod on tetraspore germination and early development of four Gracilaria species (Rhodophyta)

Ting Li

Sanya Tropical Fisheries Research Institute

Suwen Chen

South China Sea Fisheries Research Institute, Chinese Academy of Fishery Sciences

Junwei Li

South China Sea Fisheries Research Institute, Chinese Academy of Fishery Sciences

Haoxiang Hu

South China Sea Fisheries Research Institute, Chinese Academy of Fishery Sciences

Xiaoyong Xie

xyxie@scsfri.ac.cn

South China Sea Fisheries Research Institute, Chinese Academy of Fishery Sciences

Mujiao Xie

South China Sea Fisheries Research Institute, Chinese Academy of Fishery Sciences

Duyu Li

South China Sea Fisheries Research Institute, Chinese Academy of Fishery Sciences

Research Article

Keywords: Gracilaria, spore germination, salinity tolerance, light regime, seedling production, macroalgal cultivation

Posted Date: June 6th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-6783046/v1>

License:   This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Abstract

Spore-based propagation offers a sustainable alternative to vegetative reproduction in *Gracilaria* aquaculture, mitigating germplasm degradation and supporting long-term cultivation. This study examines the effects of salinity (6–36 psu), irradiance (25–115 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$), and photoperiod (8L:16D to 14L:10D) on the early development of four underutilized *Gracilaria* species native to southern China—*G. fanii*, *G. bailinae*, *G. hainanensis*, and *G. punctata*. All species were significantly influenced by salinity ($p < 0.05$), with *G. fanii* exhibited a broad salinity tolerance (18–36 psu) for tetraspore attachment and disc survival, achieving an optimal growth at 30 psu (RGR: 5.64 $\%\cdot\text{d}^{-1}$); while *G. punctata* was more sensitive to salinity extremes but showed improved survival at 30 psu (RGR: 4.75 $\%\cdot\text{d}^{-1}$). *G. bailinae* favored moderate salinity (24 psu), low irradiance (25 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$), and 14L:10D regime, achieving 76.4 % survival and 44.4 % erect thallus emergence; In contrast, *G. hainanensis* tolerated high irradiance (up to 115 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) and performed best at 30 psu under a 12L:12D regime, with 86.3% survival and 95.4% thallus emergence. These findings reveal distinct photophysiological and salinity adaptation strategies among lesser-known *Gracilaria* spp. and support their application in sustainable reproduction.

1 Introduction

The red algal genus *Gracilaria* (Rhodophyta, Gigartinales) is widely distributed in tropical and subtropical coastal waters and holds significant economic and ecological value. Industrially, *Gracilaria* serves as the principal global source of agar—a high value hydrocolloid used extensively in the food, pharmaceutical, and biotechnology sectors—while also yielding bioactive compounds with demonstrated anticancer, antioxidant, and antimicrobial properties (Armisen 1995; Bixler and Porse 2011; Rawiwan et al. 2022). Ecologically, as a rapidly growing macroalgae, *Gracilaria* functions as a biofilter through carbon sequestration and nutrient absorption in eutrophic coastal waters (Fei 2004; Spanò et al. 2022). Cultivation systems have been established in numerous countries, including China, India, the Philippines, Thailand, Brazil, and Chile (Hurtado-Ponce et al. 1992; Buschmann et al. 1995; Mantri et al. 2009; Bezerra and Marinho-Soriano 2010; Bunsom and Prathep 2012; Yang et al. 2015), contributing approximately one-third of global *Gracilaria* biomass (Mantri et al. 2009). Despite its global relevance, the *Gracilaria* aquaculture practices are facing critical constraints. Prolonged reliance on vegetative propagation has led to progressive germplasm degradation, reduced stress tolerance, and declining yields in long-term monocultures (Yang et al. 2025); Overexploitation of natural populations has accelerated the decline in resources (Mouedden et al. 2024). These threats underscore the urgent need for alternative propagation strategies to preserve genetic diversity, ensure sustainable biomass supply, and facilitate ecological restoration.

Gracilaria has a typical triphasic life cycle, alternating among haploid gametophytes and two diploid stages—carposporophytes and tetrasporophytes (Wang et al. 2010). The latter produces tetraspores through meiosis, enabling regular spore formation. This reproductive strategy enables tetraspore-based propagation to become a genetically diverse alternative to clonal methods in cultivation (Yu et al. 2013;

Zhou et al. 2016). However, the successful application of spore-based techniques hinges on a deep understanding of the early developmental stages—from spore release and substrate attachment to basal disc and erect thallus formation—all of which are highly sensitive to environmental factors such as temperature, salinity, and light (Singh and Singh 2015; Yu et al. 2016). While studies have documented environmental influences on spore development in widely cultivated species such as *G. cornea* (Orduña-Rojas and Robledo 1999), *G. pacifica* (Garza-Sánchez et al. 2000), *G. crassa* (Sukumaran and Kaliaperumal 2001), *Gracilariopsis lemaneiformis* (Ye et al. 2006; Zhou et al. 2013), *G. domingensis* (Ramlov et al. 2012), *G. chilensis* (Guillemin et al. 2013), *G. tenuistipitata* var *liui* (Yu et al. 2013), *G. gracilis* (Michetti et al. 2013; Hughes et al. 2014), *G. edulis* (K et al. 2016), and *G. fisheri* (Yang et al. 2025), there remains a substantial knowledge gap regarding the early developmental physiology of lesser-known taxa.

As the world's largest exporter of seaweeds and hydrocolloid derivatives (agar, carrageenan, and alginate) (Khan et al. 2024), China faces increasing pressure on the surging demand worldwide that now exceeds sustainable wild stock replenishment rates. Among over 30 *Gracilaria* spp. found along the Chinese coastline, commercial efforts remain disproportionately focused on a few high-yielding cultivars, notably *G. lemaneiformis* and *G. tenuistipitata* (Xu et al. 2009; Zhou et al. 2016). In contrast, species such as *G. fanii* B.M.Xia & Pan, *G. bailinae* J.Zhang & B.M.Xia, *G. hainanensis* C.F.Chang & B.M.Xia, and *G. punctata* (Okamura) Yamada remain underutilized and poorly characterized, despite their potential for aquaculture diversification and ecological deployment. Understanding their physiological responses to key environmental factors is crucial for unlocking their cultivation potential. This study aims to systematically investigate the effects of salinity, irradiance, and photoperiod on the early developmental stages of four underexplored *Gracilaria* spp. Our objectives are to: (1) elucidate the environmental regulation of early developmental processes; (2) compare interspecific differences in environmental adaptability; and (3) provide theoretical and practical foundations for optimizing spore-based seedling production protocols. The results will contribute to a deeper understanding of *Gracilaria* eco-physiology while supporting global efforts in regenerative seaweed farming and coastal ecosystem resilience.

2 Materials and Methods

2.1 Sample Collection and Preparation

Four *Gracilaria* spp. (*G. fanii*, *G. bailinae*, *G. hainanensis*, and *G. punctata*) were collected from the intertidal zone of Xuwen, Zhanjiang, China (20°16.7'N, 110°1.2'E). *G. fanii* and *G. bailinae* were sampled in January 2024, while *G. hainanensis* and *G. punctata* in April 2024. Specimens were transported in seawater-filled containers with ice packs and delivered to the laboratory in Guangzhou within 24 hours. After removal of epiphytes and debris using a soft brush, the samples were rinsed with sterilized seawater and acclimated for 3–5 days. The seawater used throughout the experiments was sourced from Qixing Bay in Shenzhen (salinity: 29.5 psu; inorganic nitrogen: 0.53 mg·L⁻¹; inorganic phosphorus:

0.04 mg·L⁻¹), filtered through fine sand, and stored in darkness for 10 days to reduce microbial load (Chen et al. 2014). Experimental salinities (6, 12, 18, 24, 30, and 36 psu) were prepared by either diluting seawater stock with distilled freshwater or enriching it with concentrated seawater, and validated using a hydrometer (Atago, PAL-07S). Average laboratory conditions corresponded with seasonal field conditions: 19.1°C and 60.5 μmol photons·m⁻²·s⁻¹ in Jan/Feb; 26.4°C and 61.7 μmol photons·m⁻²·s⁻¹ in Apr/May.

2.2 Experimental Design

2.2.1 Salinity Trial for Tetraspore Attachment

Fertile tetrasporophyte-dominant thalli were selected for tetraspores release. Algal fragments were incubated in 500 mL containers containing seawater adjusted to achieve a spore density of 10³ spores·mL⁻¹. The spore suspensions were then transferred into 500 mL beakers (8.5 cm diameter), each containing 400 mL of seawater at one of six salinity levels (6–36 psu). A sterilized microscope slide (76 × 25 mm) was placed horizontally at the bottom of each beaker as spore attachment substrates. Each salinity treatment included three replicates and was incubated under conditions described in Section 2.1. After 24 hours, spore attachment density was quantified by counting 10 randomly selected microscopic fields (40× magnification) per slide. Attachment rate (Rg) was calculated as $Rg = (S_i / S_o) \times 100\%$, where S_o is the total number of inoculated spores, and S_i is the number of attached spores. *G. hainanensis* and *G. punctata* were excluded from this trial due to consistently low attachment rates (< 1%) under all salinity conditions.

2.2.2 Salinity Trial for Tetraspore Germination

Fertile thalli were hung in aerated seawater tanks (30 × 49.5 × 30 cm) with sterilized microscope slides arranged at the bottom to collect evenly released spores. After a 3-day acclimation and spore release period, slides with uniform basal disc density were selected for testing. The initial average basal disc parameters were: *G. fanii* (1.28 ind·mm⁻², 17.80 μm), *G. bailinae* (0.49 ind·mm⁻², 20.40 μm), *G. hainanensis* (1.10 ind·mm⁻², 21.01 μm), and *G. punctata* (3.24 ind·mm⁻², 22.79 μm). Slides were transferred to 500 mL beakers (n = 3 per salinity level), each containing 400 mL seawater at the test salinities. Cultures were maintained for 30 days with weekly full water exchange and gentle daily agitation. Basal disc survival rate was calculated as $R_t = (P_t / P_o) \times 100\%$, and relative growth rate (RGR) as $RGR = [(\ln D_t - \ln D_o) / t] \times 100\%$, where P_o and P_t represent basal disc densities at day 0 and t, and D_o and D_t represent initial and final disc diameters, respectively.

2.2.3 Irradiance and Photoperiod Trial

The effects of different irradiance (25, 55, 80, 135 μmol photons·m⁻²·s⁻¹) and photoperiods (8L: 16D, 10L: 14D, 12L: 12D and 14L: 10D light: dark cycle) were evaluated using spore slides of *G. bailinae* and *G. hainanensis*. Due to limited availability of uniform slides, only these two species were included. Slides were incubated in 500 mL beakers filled with seawater under the same conditions as in Section 2.1. Light conditions were controlled in artificial light chambers that could control light (0–140 μmol

photons·m⁻²·s⁻¹) and photoperiod. Irradiance was measured using a quantum sensor (Apogee Instruments, MQ-500). Water change routine and daily maintenance were the same as in Section 2.2.2. Basal disc density and diameter were recorded at days 0 and 30. Erect thallus emergence rate (Rs) was calculated as $R_s = (U_t / P_t) \times 100\%$, where P_t is the number of basal discs and U_t is the number of erect thalli. Thallus height was measured with digital calipers (Syatek, ± 0.01 mm).

2.3 Data analysis

Data are presented as mean $\pm$ standard error (SE). One-way ANOVA followed by Duncan's post-hoc test was used to analyze the effects of salinity. Two-way ANOVA assessed the effects and interactions of irradiance and photoperiod. Levene's test was applied to verify homogeneity of variance. Significance levels were set as follows: *** $p < 0.001$ (highly significant), ** $p < 0.01$ (moderately significant), * $p < 0.05$ (significant), and $p > 0.05$ (non-significant). All statistical analyses were performed using SPSS 22.0 (IBM), and visualizations were generated with SigmaPlot 12.5 (Systat Software).

3 Results

3.1 Effects of Salinity on Tetraspore Attachment

Tetraspore attachment rates of *G. fanii* and *G. bailinae* varied significantly across salinity treatments (6–36 psu) (Fig. 1). In *G. fanii*, attachment ranged from 2.5–68.9%, with the highest value at 24 psu ($F = 12.571$, $p < 0.001$). However, attachment rates at 18–36 psu did not differ significantly ($p > 0.05$). *G. bailinae* exhibited a marked salinity-dependent increase in attachment, rising from 5.7% at 6 psu to 65.1% at 36 psu ($F = 20.388$, $p < 0.001$). Attachment rates were significantly higher at 30–36 psu compared to those at 6–18 psu ($p < 0.05$).

3.2 Salinity effects on basal disc development

Basal disc survival and RGR of the four *Gracilaria* spp. were assessed under salinity conditions ranging from 6 to 36 psu over a 30-day period (Fig. 2 & Fig. 3). In *G. fanii*, survival was significantly affected by salinity on both day 10 ($F = 5.609$, $p < 0.01$) and day 30 ($F = 7.920$, $p < 0.01$), with higher survival rates observed at 18, 24, and 30 psu. Survival rate at 6 psu declined sharply from 44.4% (day 10) to near-zero (day 30) (Fig. 2). Maximum relative growth rate (RGR, 5.64%·d⁻¹) occurred at 30 psu ($F = 83.918$, $p < 0.001$), significantly exceeding other salinities (Fig. 3). For *G. bailinae*, survival rates differed significantly at 10 days ($F = 20.978$, $p < 0.001$) and 30 days ($F = 19.391$, $p < 0.001$), showed significantly higher survival at 18 psu on day 10 (76.4%) and at 24 psu on day 30 (53.4%) (Fig. 2). The highest RGR (4.35%·d⁻¹) occurred at 24 psu, which was significantly higher than other treatments ($F = 17.597$, $p < 0.001$), except for 30 psu (Fig. 3). *G. hainanensis* also exhibited significant survival variations throughout the experiment (10 days: $F = 6.978$, $p < 0.01$; 30 days: $F = 6.738$, $p < 0.01$), with the highest survival rate consistently presented at 30 psu (Fig. 2). However, RGR did not differ significantly across treatments ($p > 0.05$), with an average of 2.73%·d⁻¹ (Fig. 3). *G. punctata* demonstrated poor survival across salinities except at 30 psu where survival maintained above 15%, with significant differences at 10 days ($F =$

67.532, $p < 0.001$) and 30 days ($F = 19.506$, $p < 0.001$) (Fig. 2). Higher RGR values ($4.08\text{--}5.07\%\cdot\text{d}^{-1}$) occurred at 18–36 psu ($F = 7.729$, $p < 0.01$), with no significant differences among these salinities. (Fig. 3). Both *G. bailinae* and *G. punctata* showed negligible survival at 6 psu throughout the observation period (Fig. 2).

3.3 Effects of Irradiance and Photoperiod on *G. bailinae*

Two-way ANOVA revealed that both irradiance ($p < 0.001$) and photoperiod ($p < 0.01$) significantly influenced basal disc survival and RGR, though their interaction was not significant ($p > 0.05$) (Table 1). Survival rate was highest (76.4%) under low irradiance ($25\text{ }\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) combined with an extended photoperiod (14L: 10D) (Fig. 4). Survival at this irradiance was 1.9×, 2.4×, and 2.6× higher than at 55, 85, and 105 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, respectively. Peak RGR was also observed at 25 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, with values 25–60% higher than at elevated irradiance levels (55–105 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$). Under low irradiance (25 and 55 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$), RGR increased with the lengthening of photoperiod. Erect thallus emergence was significantly affected by irradiance ($p < 0.001$) but not by photoperiod ($p > 0.05$) (Table 1). The highest emergence rate (44.4%) occurred under 25 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ and 14L: 10D, with emergence sharply reduced at higher light intensities (Fig. 4). At 115 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, erect thalli only emerged under short light period (8L: 16D).

Table 1
Two-way ANOVA results for effects of irradiance and photoperiod on survival rate (%), RGR (%·d⁻¹), and erect thallus emergence rate (%) of *G. bailinae* basal discs.

Source of variation	<i>df</i>	<i>MS</i>	<i>F</i>	<i>p</i>
Survival rate %				
Irradiance (I)	3	0.416	9.226	< 0.001
Photoperiod (P)	3	0.272	6.043	0.002
I × P	9	0.066	1.467	0.202
Error	32	0.045		
RGR %·d ⁻¹				
Irradiance (I)	3	3.023	15.215	< 0.001
Photoperiod (P)	3	1.164	5.858	0.003
I × P	9	0.4	2.014	0.07
Error	32	0.199		
erect thallus emergence rate %				
Irradiance (I)	3	0.184	21.957	< 0.001
Photoperiod (P)	3	0.019	2.242	0.102
I × P	9	0.016	1.939	0.082
Error	32	0.008		

3.4 Effects of Irradiance and Photoperiod on *G. hainanensis*

Two-way ANOVA showed that both irradiance ($p < 0.001$) and photoperiod ($p < 0.001$) significantly affected basal disc survival, with a notable interaction effect ($p < 0.001$) (Table 2). Survival increased from 22.4% at 25 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ to 54.3% at 105 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ (Fig. 5). Significant differences were observed under 10L: 14D ($p < 0.01$) and 12L: 12D ($p < 0.05$) but not under 14L: 10D ($p > 0.05$). Both RGR and erect thallus emergence increased with irradiance ($p < 0.001$) and photoperiod ($p < 0.01$), though no significant interaction was found ($p > 0.05$) (Table 2). Among irradiances, RGR ranged from 1.44%·d⁻¹ to 4.57%·d⁻¹ (peaking at 115 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$), while thallus emergence ranged from 11.1–80.3% (maximum at 85 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) (Fig. 5). At 25 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, no upright thalli emerged under 10L: 14D and 12L: 12D. Upright thallus height demonstrated significant responses to irradiance ($p < 0.001$), photoperiod ($p < 0.01$), and their interaction ($p < 0.05$), with the tallest thalli (2.34 mm) observed under 85 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ and 14L: 10D (Table 2, Fig. 5).

Table 2
Two-way ANOVA results for effects of irradiance and photoperiod on survival rate (%), RGR (%·d⁻¹), erect thallus emergence rate (%), and height of erect thallus (mm) of *G. hainanensis* basal discs.

Source of variation	<i>df</i>	<i>MS</i>	<i>F</i>	<i>p</i>
Survival rate %				
Irradiance (I)	3	0.18	12.156	< 0.001
Photoperiod (P)	2	0.527	35.566	< 0.001
I × P	6	0.062	4.194	0.005
Error	24	0.015		
RGR % d ⁻¹				
Irradiance (I)	3	19.318	39.478	< 0.001
Photoperiod (P)	2	3.001	6.133	0.007
I × P	6	1.017	2.079	0.094
Error	24	0.489		
erect thallus emergence rate %				
Irradiance (I)	3	0.942	24.651	< 0.001
Photoperiod (P)	2	0.323	8.439	0.002
I × P	6	0.084	2.2	0.078
Error	24	0.038		
height of erect thallus mm				
Irradiance (I)	3	8.321	31.237	< 0.001
Photoperiod (P)	2	2.309	8.666	0.001
I × P	6	0.782	2.937	0.027
Error	24	0.266		

4 Discussion

4.1 Salinity Regulation Effect

Salinity is a fundamental abiotic factor influencing physiology, distribution, and viability of marine macroalgae by modulating osmotic balance and membrane permeability (Yokoya et al. 1999; Raikar et al. 2001). Most *Gracilaria* spp. inhabit intertidal or estuarine zones, where salinity fluctuates significantly

due to freshwater input (Barbosa-Silva et al. 2023). Understanding salinity tolerance in *Gracilaria* spp. is essential for successful cultivation, especially during early developmental transitions such as spore germination and basal disc formation. Our findings reveal clear interspecific differences in salinity preferences of the four tested *Gracilaria* spp., reflecting their respective ecological niches and adaptive strategies. *G. fanii* exhibited a broad salinity tolerance (18–36 psu) for both tetraspore attachment and basal disc survival, with peak growth at 30 psu, consistent with the natural seawater salinity along exposed coastlines. It also aligns with reports for *G. fisheri* and *G. lemaneiformis*, favoring higher salinities (30–35 psu) (Zhou et al. 2013; Yang et al. 2025), but diverging from species like *G. edulis* and *G. tenuistipitata*, which prefers low-salinity conditions (15 psu) (Yu et al. 2013; K et al. 2016). Conversely, *G. bailinae* achieved maximal survival and growth at moderate salinity (24 psu), suggesting adaptation to brackish water or estuarine zones where salinity fluctuates but remains below oceanic levels. Both *G. hainanensis* and *G. punctata* showed optimal survival narrowly around 30 psu, and exhibited steep declines in both low and high extremes. These patterns suggest stenohaline behavior and potential vulnerability to salinity shifts induced by freshwater inflow or climate change. Under hyposaline conditions (6–12 psu), cytoplasmic shrinkage and bleaching were observed, likely due to osmotic stress disrupting cell wall integrity and metabolic regulation, as similarly reported in *G. domingensis* and *G. edulis* (Ramlov et al. 2012; Yu et al. 2013). These results collectively define species-specific osmoregulatory thresholds in the early developmental stages of *Gracilaria* spp., providing critical parameters to optimize seedling culture protocols and minimize early-stage mortality. Understanding such physiological windows is essential for habitat matching in aquaculture site selection and ecological restoration.

4.2 Light-Driven Regulation Effect

Light is not only the primary energy source for photosynthesis but also a key developmental cue that influences morphogenesis, biological rhythm, and reproductive transitions in red algae (Fortes and Lüning 1980; Ye et al. 2006; Jung and Kim 2025). Light parameters include irradiance (light intensity) and photoperiod (light duration). Although current research on early developmental photoresponses in *Gracilaria* spp. have primarily concentrated on the effects of irradiance, photoperiod regulation has been considerably understudied, as well as the interreaction between irradiance and photoperiod (Zhou et al. 2013; Michetti et al. 2013). In this study, the two species evaluated under varied light regimes, *G. bailinae* and *G. hainanensis*, displayed contrasting photoadaptive strategies. *G. bailinae* displayed photoinhibition under elevated irradiance ($> 55 \mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$), while thriving under low light ($25 \mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) combined with an extended photoperiod (14L:10D). This strategy is consistent with other shade-adapted taxa such as *G. lemaneiformis* which were preferred in $30 \mu\text{mol m}^{-2} \text{ s}^{-1}$ (Ye et al. 2006), indicating a high photosynthetic efficiency at low light intensities but susceptibility to oxidative stress at higher irradiance. In contrast, *G. hainanensis* exhibited improved survival and growth with increasing irradiance up to $115 \mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ under 12L: 12D regime. This high irradiance tolerance suggests the effective of photoprotective pigments and efficient high-light acclimation mechanisms, comparable to those reported in *G. edulis* and *G. tenuistipitata* var. *liui* which exhibited optimal growth at around $100 \mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ (Yu et al. 2013). The broad range of light optima

reported across *Gracilaria* spp.—spanning from 15 to 200 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ (Ramlov et al. 2012; Miranda et al. 2012; Zhou et al. 2013)—reflects their considerable photophysiological plasticity. Notably, the significant interaction between irradiance and photoperiod observed in *G. hainanensis* for survival and thallus development suggests a synergistic control of morphogenesis. Here, extended photoperiods may compensate for suboptimal light intensity by increasing the total daily photon flux or may lead to accumulated photoinhibition damage under high light intensity.

4.3 Practical Implications

Coastal habitat degradation and overharvesting have led to alarming declines in natural *Gracilaria* populations, threatening both biodiversity and sustainable industry supply chains (Wang et al. 2010; Irwani 2013). While traditional cultivation relies heavily on a few species, such as *G. lemaneiformis*, broader germplasm development remains limited. Through long-term field surveys and systematic sampling of macroalgae along southern China's coastline, we identified four native *Gracilaria* spp. exhibiting the capability of substantial tetraspore release after stripped from their natural environment. This study successfully established spore-based seedling cultures for all four species and cultivated erect thallus in *G. bailinae* and *G. hainanensis*, contributes to the genetic rejuvenation of seeding stocks and species diversification in *Gracilaria* cultivation. The observed physiological resilience of *G. bailinae* under low light and moderate salinity positions it as a strong candidate for indoor or shaded culture systems where energy input for lighting can be minimized. On the other hand, *G. hainanensis*—though requiring high irradiance—demonstrates superior erect thallus formation and rapid growth, making it suitable for outdoor cultivation in high-transparency environments or integrated multitrophic aquaculture (IMTA) settings. Such propagation systems may also aid coastal restoration by providing seed banks for re-establishing macroalgal beds in nutrient-rich coastline or estuarine zones. While optimal conditions for *G. fanii* and *G. punctata* remain partially undefined due to sample limitations and the rarity of specimens, their salinity responses provide a basis for future work.

Declarations

Funding This research was funded by Finance science and technology project of hainan province/Hainan Provincial Natural Science Foundation (323QN329) and Shenzhen Science and Technology Program (KCXST20221021111206015).

Competing interest The authors declare no conflicts of interest, financial or otherwise, that may have influenced the outcome of this study.

Data availability Data are available from the corresponding authors upon reasonable request.

Code availability Not applicable

Author contribution Li T: conceptualization, funding acquisition, experiment implement, data analysis, writing and editing; Chen SW: methodology design and materials collection; Li JW: review and editing; Hu HX: experiment implement and data collection; Xie XX: administrative support, review and editing; Xie M

J: technical and logistic support; Li DY: sample collection. All authors contributed to the article and approved the submitted version.

References

1. Armisen R (1995) World-wide use and importance of *Gracilaria*. J Appl Phycol 7:231–243.
2. Barbosa-Silva MS, Borburema HDS, Fernandes FDO, de Nóbrega MF, Marinho-Soriano E (2023) Projecting environmental suitability areas for the seaweed *Gracilaria birdiae* (Rhodophyta) in Brazil: Implications for the aquaculture pertaining to five environmentally crucial parameters. J Appl Phycol 35:773–784.
3. Bezerra AF, Marinho-Soriano E (2010) Cultivation of the red seaweed *Gracilaria birdiae* (Gracilariales, Rhodophyta) in tropical waters of northeast Brazil. Biomass and Bioenergy
4. Bixler HJ, Porse H (2011) A decade of change in the seaweed hydrocolloids industry. J Appl Phycol 23:321–335.
5. Bunsom C, Prathep A (2012) Effects of salinity, light intensity and sediment on growth, pigments, agar production and reproduction in *Gracilaria tenuistipitata* from Songkhla Lagoon in Thailand. Phycological Research 60:169–178.
6. Buschmann AH, Westermeier R, Retamales CA (1995) Cultivation of *Gracilaria* on the sea-bottom in southern Chile: a review. J Appl Phycol 7:291–301.
7. Chen SW, Chen LX, Zhu CB, Guo YJ, Guo YH (2014) Spore Culture of Red Seaweed, *Gloiopeltis furcata*. J World Aquacult Soc 45:487–492.
8. Fei X (2004) Solving the coastal eutrophication problem by large scale seaweed cultivation. Hydrobiologia 512:145–151.
9. Fortes MD, Lüning K (1980) Growth rates of North Sea macroalgae in relation to temperature, irradiance and photoperiod. Helgolander Meeresunters 34:15–29.
10. Garza-Sánchez F, Zertuche-González JA, Chapman DJ (2000) Effect of Temperature and Irradiance on the Release, Attachment and Survival of Spores of *Gracilaria pacifica* Abbott (Rhodophyta). Botanica Marina 43:205–212.
11. Guillemín M-L, Sepúlveda RD, Correa JA, Destombe C (2013) Differential ecological responses to environmental stress in the life history phases of the isomorphic red alga *Gracilaria chilensis* (Rhodophyta). J Appl Phycol 25:215–224.
12. Hughes MH, Michetti KM, Leonardi PI (2014) Settlement of *Gracilaria gracilis* carpospores (Gracilariales, Rhodophyta) on natural substrates from the southwestern Atlantic coast (Chubut, Argentina). Botanica Marina 57:131–137.
13. Hurtado-Ponce AQ, Samonte GPb, Luhan MaR, Guanzon N (1992) *Gracilaria* (Rhodophyta) farming in Panay, Western Visayas, Philippines. Aquaculture 105:233–240.
14. Irwani I (2013) Spores Setting of *Gracilaria gigas* (Rhodophyta ,Gracilariales) in Different Artificial Substrates. Buletin Oseanografi Marina 2:48–50.

15. Jung JW, Kim JK (2025) Effects of light quality and photoperiod on growth, photosynthesis and pigments of tetrasporophytes of *Asparagopsis taxiformis* (Bonnemaisoniaceae, Rhodophyta). J Appl Phycol 37:1289–1296.
16. Ashok KS, Yadav S, Saminathan KR, Monisha N, Malarvizhi J, Ganesan M, Mantri VA (2016) An orthogonal design to optimize seed production, out-planting, and cultivation of the industrially overexploited red alga *Gracilaria edulis* (Rhodophyta). J Appl Phycol 28:1215–1223
17. Khan N, Sudhakar K, Mamat R (2024) Macroalgae farming for sustainable future: Navigating opportunities and driving innovation. Heliyon 10:e28208.
18. Mantri VA, Thakur MC, Kumar M, Reddy CRK, Jha B (2009) The carpospore culture of industrially important red alga *Gracilaria dura* (Gracilariales, Rhodophyta). Aquaculture 297:85–90.
19. Michetti KM, Martín LA, Leonardi PI (2013) Carpospore release and sporeling development in *Gracilaria gracilis* (Gracilariales, Rhodophyta) from the southwestern Atlantic coast (Chubut, Argentina). J Appl Phycol 25:1917–1924.
20. de Miranda GEC, Yokoya NS, Fujii MT (2012) Effects of temperature, salinity and irradiance on carposporeling development of *Hidropuntia caudata* (Gracilariales, Rhodophyta). Rev bras farmacogn 22:818–824.
21. Mouedden R, Abdellaoui S, El Madani F, El Ouamari N, Slimani D, Kasmi K, Taibi M, Zahir I, Chaabane K (2024) *Gracilaria gracilis* - A Review of Ecological Knowledge, Chemical Composition, Cultivation, and Applications. Ecological Engineering & Environmental Technology (EEET) 25:276–287
22. Orduña-Rojas J, Robledo D (1999) Effects of Irradiance and Temperature on the Release and Growth of Carpospores from *Gracilaria cornea* J. Agardh (Gracilariales, Rhodophyta). 42:315–319.
23. Raikar S, Iima M, Fujita Y (2001) Effect of temperature, salinity and light intensity on the growth of *Gracilaria* spp. (Gracilariales, Rhodophyta) from Japan, Malaysia and India. Indian Journal of Marine Sciences 30:98–104
24. Ramlov F, de Souza JMC, Farias A, Maraschin M, Horta PA, Yokoya NS (2012) Effects of temperature, salinity, irradiance, and nutrients on the development of carposporelings and tetrasporophytes in *Gracilaria domingensis* (Kütz.) Sonder ex Dickie (Rhodophyta, Gracilariales). Botanica Marina 55:253–259.
25. Rawiwan P, Peng Y, Paramayuda IGPB, Quek SY (2022) Red seaweed: A promising alternative protein source for global food sustainability. Trends in Food Science & Technology 123:37–56.
26. Singh SP, Singh P (2015) Effect of temperature and light on the growth of algae species: A review. Renewable and Sustainable Energy Reviews 50:431–444.
27. Spanò N, Di Paola D, Davide, Albano M, Manganaro MS, D'Iglio C, Capillo G, Savoca S (2022) Growth performance and bioremediation potential of *Gracilaria gracilis* (Steentoft, L.M. Irvine & Farnham, 1995). International Journal of Environmental Studies 79:748–760.
28. Sukumaran S, Kaliaperumal N (2001) Sporulation in *Gracilaria crassa* Harvey ex. J. Agardh at different environmental factors. Seaweed Research & Utilisation 23:81–87

29. Wang Z, Wang G, Niu J, Wang W, Peng G (2010) Optimization of conditions for tetraspore release and assessment of photosynthetic activities for different generation branches of *Gracilaria lemaneiformis* Bory. Chin J Ocean Limnol 28:738–748.
30. Xu Y, Wei W, Fang J (2009) Effects of salinity, light and temperature on growth rates of two species of *Gracilaria* (Rhodophyta). Chin J Ocean Limnol 27:350–355.
31. Yang W, Lin K, Zeng Z, et al (2025) Optimizing temperature, light intensity, and salinity for carpospore culture of *Gracilaria fisheri*. Aquaculture Reports 42:102749.
32. Yang Y, Chai Z, Wang Q, Chen W, He Z, Jian S (2015) Cultivation of seaweed *Gracilaria* in Chinese coastal waters and its contribution to environmental improvements. Algal Research 9:236–244.
33. Ye N, Wang H, Wang G (2006) Formation and early development of tetraspores of *Gracilaria lemaneiformis* (*Gracilaria*, Gracilariaceae) under laboratory conditions. Aquaculture 254:219–226.
34. Yokoya NS, Kakita H, Obika H, Kitamura T (1999) Effects of environmental factors and plant growth regulators on growth of the red alga *Gracilaria vermiculophylla* from Shikoku Island, Japan. Hydrobiologia 398:339–347.
35. Yu C-H, Lim P-E, Phang S-M (2013) Effects of irradiance and salinity on the growth of carpospore-derived tetrasporophytes of *Gracilaria edulis* and *Gracilaria tenuistipitata* var liui (Rhodophyta). J Appl Phycol 25:787–794.
36. Yu R, Lin A, Xie X, Wang H, Zhang F, Liu G, Wang G, Lin C (2016) Importance of coalescent development of *Gracilaria* sp. tetraspores. Phycological Research 64:200–203.
37. Zhou W, Sui Z, Wang J, Hu Y, Kang KH, Kim HB, Niaz Z (2016) Mass cultivation of economically important red alga *Gracilariopsis lemaneiformis* (Gracilariaceae, Rhodophyta) from tetraspores and carpospores. Aquaculture 460:25–31.
38. Zhou W, Sui Z, Wang J, Chang L (2013) An orthogonal design for optimization of growth conditions for all life history stages of *Gracilariopsis lemaneiformis* (Rhodophyta). Aquaculture 392–395:98–105.

Figures

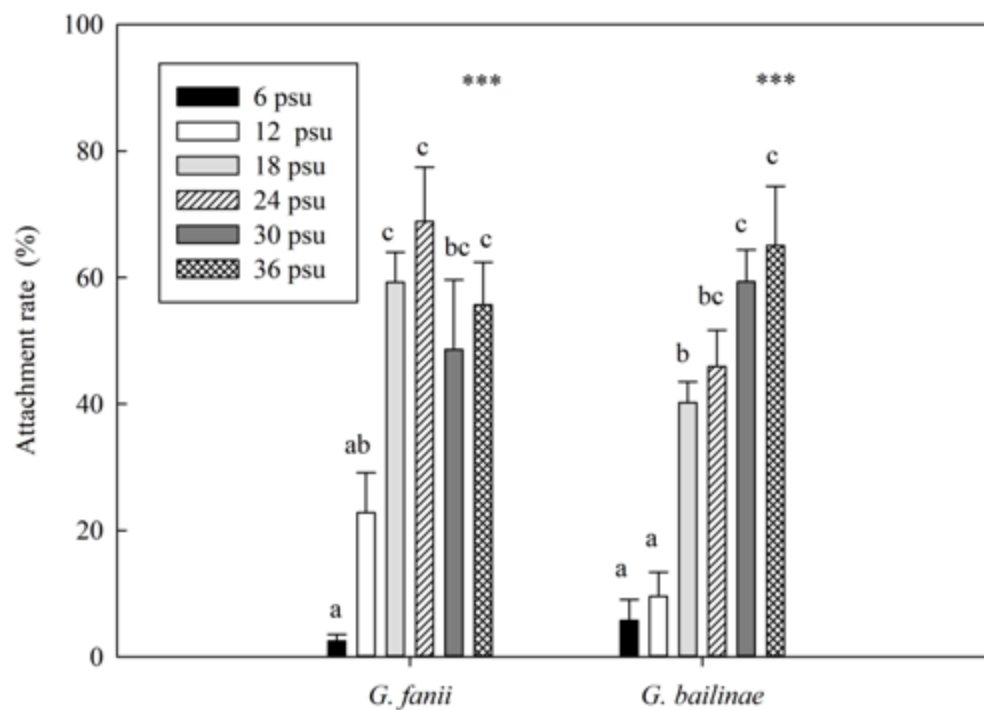


Figure 1

Tetraspores attachment rates (%) of *G. fanii* and *G. bailinae* under varying salinity levels (6–36 psu). Data are presented as mean $\pm$ SE ($n = 3$). Different lowercase letters denote statistically significant differences among treatments ($p < 0.05$).

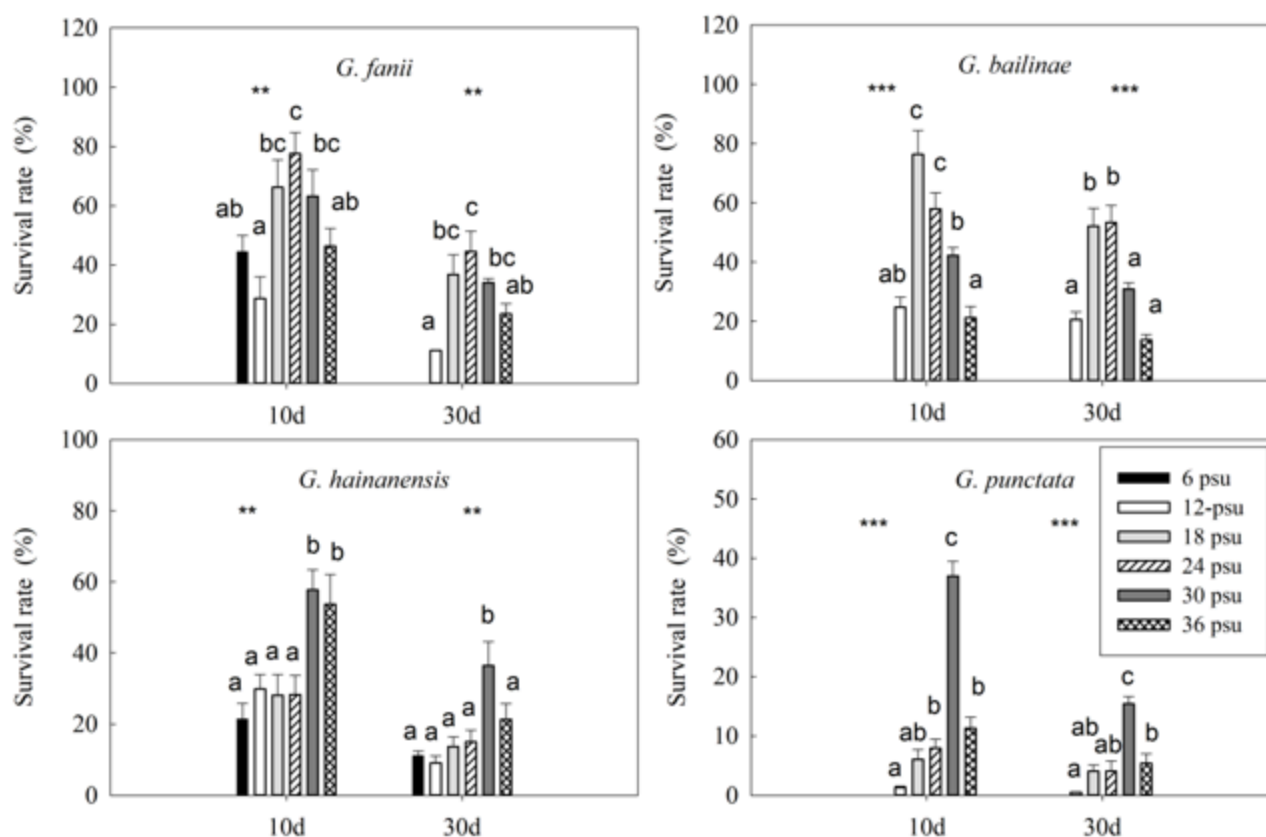


Figure 2

Basal disc survival rate (%) of four *Gracilaria* spp. under varying salinity levels (6–36 psu) over a 30-day culture period. Data are presented as mean $\pm$ SE ($n = 3$). Different lowercase letters denote statistically significant differences among treatments ($p < 0.05$).

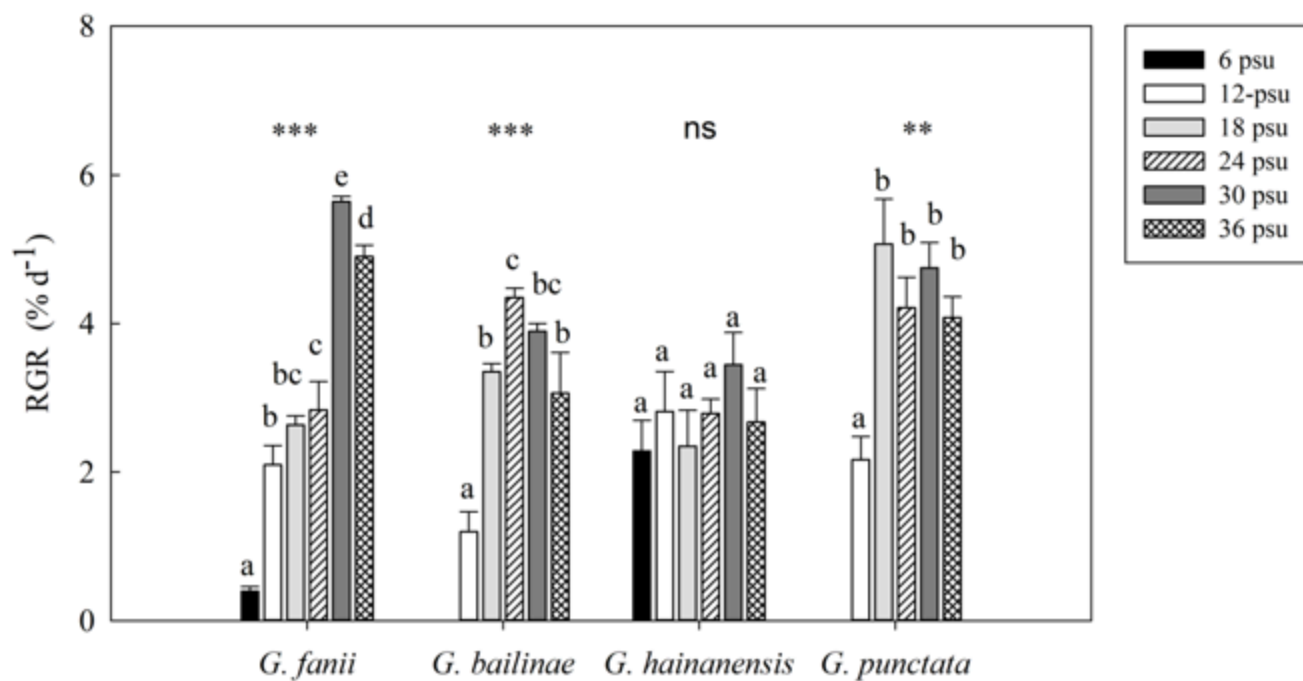


Figure 3

Relative growth rates (RGR, %·d⁻¹) of basal discs for four *Gracilaria* spp. under varying salinity levels (6–36 psu). Data are presented as mean ± SE (n = 3). Different lowercase letters denote statistically significant differences among treatments (p < 0.05).

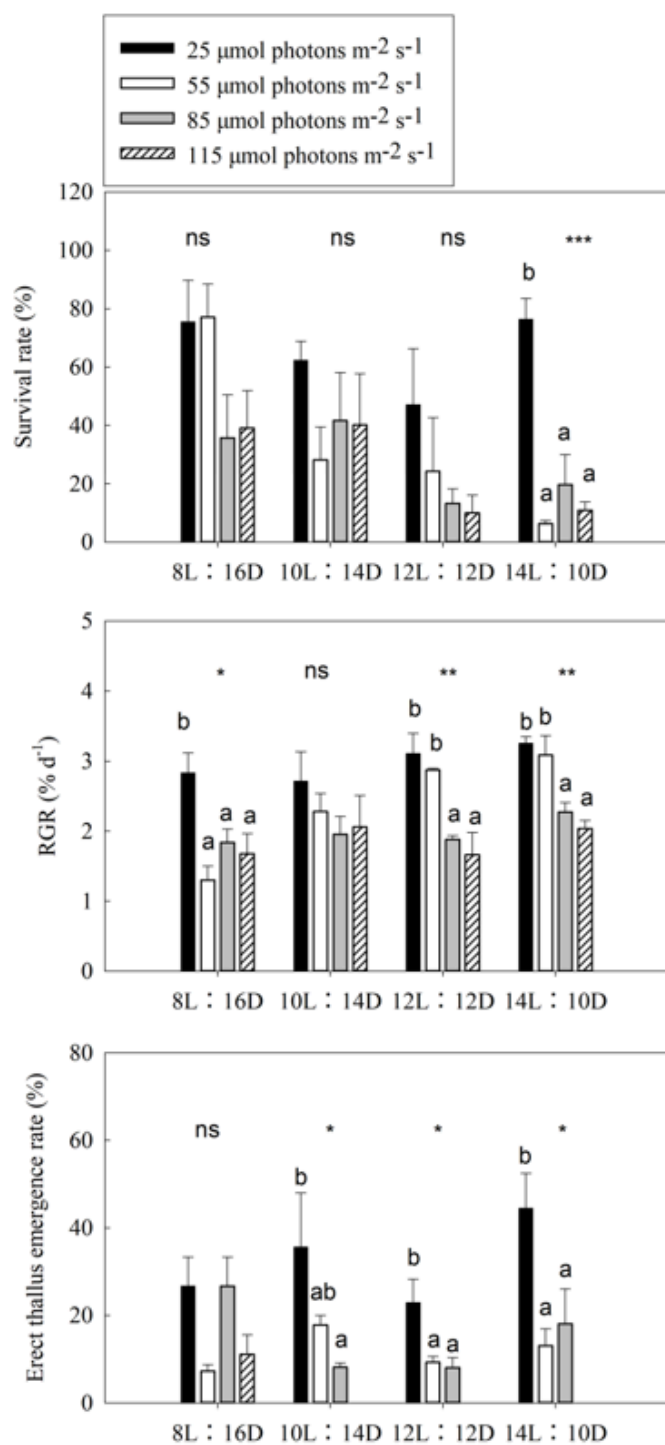


Figure 4

Survival rate (%), RGR (% $\cdot$ d⁻¹), and erect thallus emergence rate (%) of *G. bailinae* under four irradiance levels (25–115 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) and four photoperiods (8L: 16D to 14L: 10D). Data are presented as mean $\pm$ SE ($n = 3$). Different lowercase letters denote statistically significant differences among treatments ($p < 0.05$).

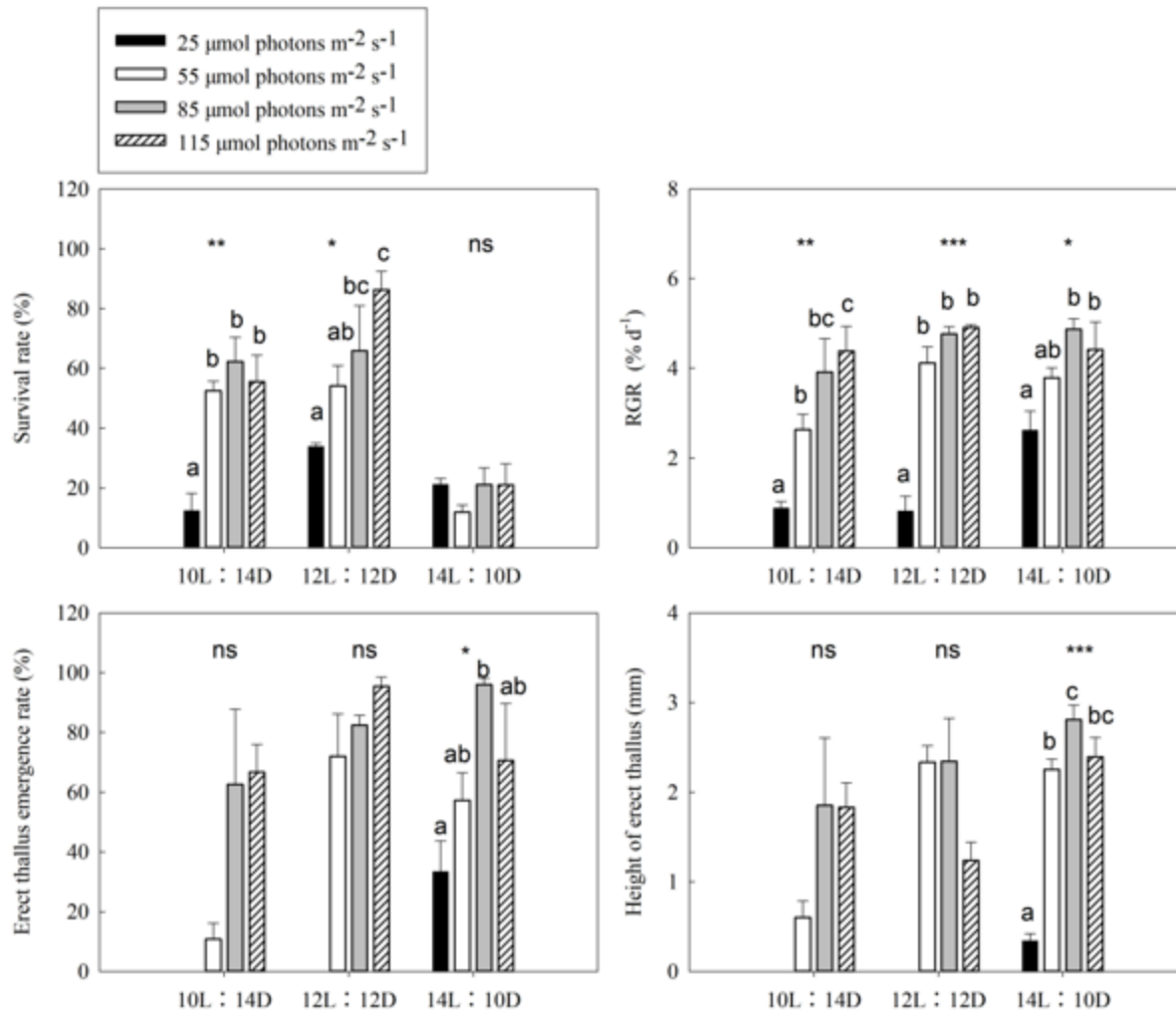


Figure 5

Survival rate (%), RGR (% d^{-1}), erect thallus emergence rate (%), and height of erect thallus (mm) of *G. hainanensis* under four irradiance levels (25–115 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$) and three photoperiods (10L: 14D to 14L: 10D). Data are presented as mean $\pm$ SE ($n = 3$). Different lowercase letters denote statistically significant differences among treatments ($p < 0.05$).